

S M Abdulla Al Mamun

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Education

The University of Manchester <i>Master of Science (M.Sc.) in Artificial Intelligence</i>	Manchester, UK 2025 – 2026 (Ongoing)
BRAC University <i>Master of Science (M.Sc.) in Computer Science and Engineering — CGPA: 4.00/4.00</i>	Dhaka, Bangladesh 2025
BRAC University <i>Bachelor of Science (B.Sc.) in Computer Science and Engineering — CGPA: 3.70/4.00</i>	Dhaka, Bangladesh 2021

Research Experience

Information Management Group (IMG), The University of Manchester <i>MSc Dissertation Researcher — Dynamic Resource Management for Intelligent Transportation Systems</i> <i>Present</i>	Manchester, UK 2026 –
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- Researching resource-aware decision-making in vehicular edge computing (VEC), where tasks must be offloaded, placed, admitted, and executed across finite roadside infrastructure under changing traffic, workload, and communication conditions.
- Developed **TrafficTwin**, an auditable research platform integrating traffic simulation and edge-computing experiments to evaluate resource-allocation policies with reproducible lifecycle, latency, deadline-attainment, and rejection accounting.
- Evaluated multi-agent reinforcement learning and deterministic scheduling/placement strategies for task execution across roadside units (RSUs), including constrained admission, forwarding, and load-distribution mechanisms.
- Built reproducible data and evidence pipelines for Manchester-oriented traffic scenarios, combining SUMO-based experimentation with real-world traffic-data integration and exact provenance for research validation.
- Supervised by **Dr Sandra Sampaio**; research focus: dynamic resource management, intelligent transportation systems, vehicular edge computing, and resource-aware applied AI/ML.

Computer Vision and Intelligent Systems (CVIS) Lab, BRAC University <i>Research Assistant</i>	Dhaka, Bangladesh 2021 – 2023
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- Led the end-to-end research lifecycle for “An Interpretable Transformer-Based Approach to Classify Malaria from Blood Cell Images,” from literature review and experimental design to manuscript preparation and peer-reviewed publication.
- Developed and benchmarked advanced deep learning frameworks (Vision Transformer, Swin Transformer, VGG16), optimizing model hyperparameters to achieve a state-of-the-art classification accuracy of 96.04%.
- Architected a robust, scalable data processing pipeline for 27,558 medical images, implementing strict protocols for normalization, geometric augmentation, and train-validation-test splitting to ensure model generalizability.
- Integrated Explainable AI (XAI) methodologies, utilizing Grad-CAM to generate saliency maps that visualized the model’s attention mechanisms, bridging the gap between black-box AI and clinical interpretability.
- Collaborated with multidisciplinary academic teams to synthesize research findings, presenting methodology and results at international conferences.

Intelligent Systems for Health <i>Researcher</i>	Dhaka, Bangladesh 2023 – 2025
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- Assisted in publications related to efficient healthcare systems in Bangladesh.
- Authored 2 academic papers pertaining to morbid diseases.
- Developed a Dialectical Behavior Therapy (DBT) app with AI integration.

Publications

- S. M. Dipto, M. Islam, S. M. A. Al Mamun, M. K. Hossain, A. M. Esfar-E.-Alam and M. A. Alam, “An Interpretable Transformer-Based Approach to Classify Malaria from Blood Cell Images,” in *2024 International Conference on Sustainable Technology and Engineering (i-COSTE)*, Perth, Australia, 2024, pp. 1–6.

Teaching Experience

Selim Reza Habib Degree College <i>Lecturer, Department of ICT</i>	Pabna, Bangladesh Jun 2023 – Dec 2023
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- Taught ICT for higher secondary school students; prepared questions and effective modules.

Professional Experience

ByteWave

Machine Learning Engineer

Dhaka, Bangladesh

2023 – 2025

- Engineered and deployed a time-series demand forecasting model (Prophet, XGBoost) to predict SKU-level inventory needs, minimizing warehouse holding costs and preventing stockouts for international imports.
- Developed a predictive supply chain model to estimate the actual arrival time of overseas shipping containers, significantly improving the accuracy of customer-facing restock notifications.
- Designed and integrated a collaborative filtering recommendation system that served personalized product suggestions via REST APIs, increasing the platform's average order value.

ByteWave

Backend Software Engineer

Dhaka, Bangladesh

2022 – 2023

- Architected robust backend APIs in Python (FastAPI) to support e-commerce operations, laying the data infrastructure foundation for future machine learning pipelines.
- Built a dynamic pricing engine that automatically adjusted product prices based on real-time warehouse inventory levels and shipping container costs.

Lumina Global

Java Backend Developer

Dhaka, Bangladesh

2019 – 2020

- Engineered scalable RESTful APIs using Java and Spring Boot to power real-time order tracking and inventory management modules, processing over 15,000 daily supply chain transactions.
- Integrated external carrier APIs and AWS S3 to automate cloud storage, securely processing and archiving over 2,000 digital invoices and shipment documents weekly.
- Optimized backend database interactions using Spring Data JPA and PostgreSQL, reducing query latency by 35% on complex multi-warehouse inventory search queries.
- Authored comprehensive unit and integration tests using JUnit and Mockito, achieving 85% code coverage across core order-processing microservices and significantly minimizing production defects.

Projects

MasalaEngine

Lead Developer

Academic Project

2026

- Led the development of a highly accurate Retrieval-Augmented Generation (RAG) pipeline for a specialized culinary corpus, achieving 95.20% Recall@5 and 0.857 nDCG@5.
- Implemented a hybrid retrieval architecture combining dense vector search (FAISS, BGE embeddings) with sparse keyword search (BM25), optimized by a Cross-Encoder re-ranker.
- Evaluated and benchmarked multiple token-chunking strategies (recursive vs. fixed-size) to maximize context retention within strict 250-token budgets.
- Engineered strict, grounded prompts with deterministic decoding to prevent LLM hallucinations, validating output quality using BERTScore and lexical semantic similarity metrics.

PaperStack (Literature IDE)

Creator & Lead Developer

Personal Project

2025 – 2026

- Engineered a full-stack, multimodal RAG platform (FastAPI, React, PostgreSQL) designed as an interactive “Literature IDE” for querying and reasoning over ML/AI research papers.
- Architected an advanced document ingestion pipeline utilizing PyMuPDF and GPT-5.5 vision models to accurately extract structured claims, empirical metrics, and table/figure data.
- Developed a hybrid search engine combining pgvector semantic search with full-text keyword search, fused via Reciprocal Rank Fusion (RRF) and optimized by LLM listwise re-ranking.
- Integrated GraphRAG capabilities and an interactive D3-force network graph to map out conceptual relationships, citations, and dataset performance comparisons across the library.

SkillSight UK

Machine Learning Engineer

Personal Project

2026

- Developed an automated NLP pipeline to extract a 13-category skill taxonomy from UK data/business analyst job postings, evaluating against a manually annotated gold-standard corpus.

- Architected a rigorous evaluation framework incorporating stratified nested cross-validation, bootstrapping, and inductive TF-IDF scoring to systematically eliminate data leakage.
- Engineered and benchmarked multiple lexical baselines (cosine similarity vs. weighted lexical TF-IDF), achieving a peak 0.938 Macro-F1 score on the internal holdout set.

Technical Skills

- **Programming & Development:** Java, Python, JavaScript, MERN Stack, Spring Boot, Laravel, MySQL, MariaDB, MongoDB, MASM, PHP.
- **Software & Analysis:** TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, R, SPSS, STATA, Tableau, MATLAB, Mathematica, MS Office Suite.
- **Interpersonal:** Project Management, Leadership, Negotiation, Public Speaking, Cultural Awareness.
- **Languages:** English (Fluent, PTE Score: 90/90), Bengali (Native).

Grants, Awards, and Scholarships

- National Science and Technology (NST) Fellowship — Ministry of Science and Technology, Government of Bangladesh, 2026
- Commonwealth Masters Scholarship, UK — 2025
- Science and Technology Fellowship (MS abroad) — Science and Technology Fellowship Trust, Ministry of Science and Technology, 2024
- Merit Based Scholarship, BRAC University — 2018
- Secondary School Certificate Scholarship, Dhaka — 2014
- Junior School Certificate Scholarship, Dhaka — 2011

Volunteering & Activities

- Robotics Club, BRAC University: Member of the Mongol Tori team (2018 – 2022).
- Computer Club, BRAC University: Programming Mentor.
- Mathematics and Natural Sciences Club: Member (2018 – 2022).
- Community Service: Community Teacher, Residential Community.